Circuit Simulation Lab:9
S-R Flip-Flop

Introduction

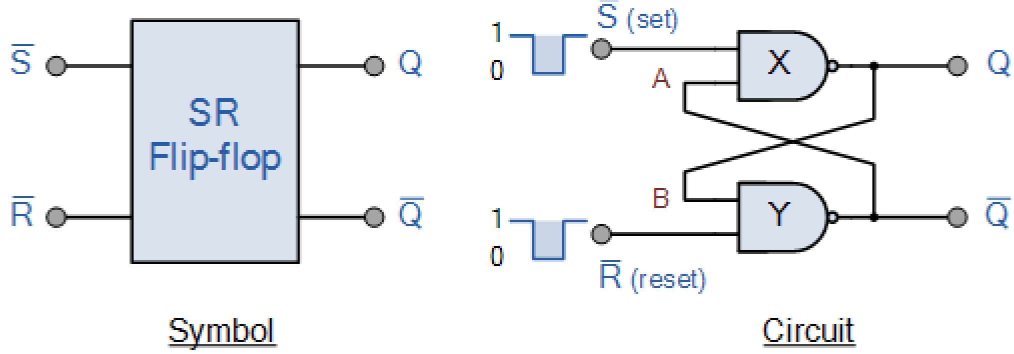
The purpose of this experiment is to introduce the design of simple sequential circuits, in this case S-R Flip Flop. The SR flip flop is a 1-bit memory bistable device having two inputs, i.e., SET and RESET. The SET input 'S' set the device or produce the output 1, and the RESET input 'R' reset the device or produce the output 0. The SET and RESET inputs are labeled as S and R, respectively. It is a single bit storage element. It has only two logic gates. The output of each gate is connected to the input of another gate.

Software tools Requirement

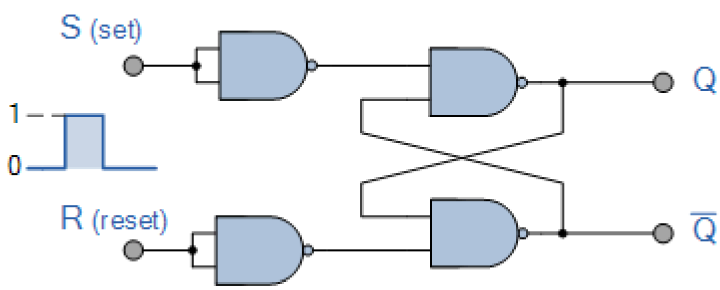
Modelsim (Siemens)

Xilinx Vivado

Logic Diagram

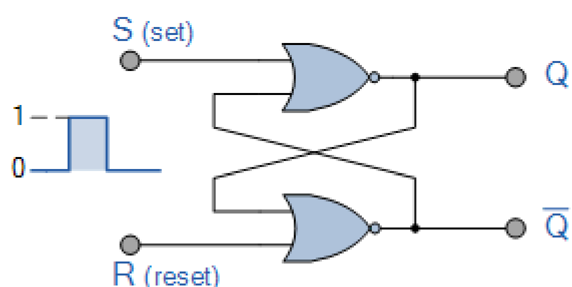


State	S	R	Q	$\overline{Q}$	Description
Set	1	0	0	1	Set $\overline{Q}$ » 1
	1	1	0	1	no change
Reset	0	1	1	0	Reset $\overline{Q}$ » 0
	1	1	1	0	no change
Invalid	0	0	1	1	Invalid Condition



NAND Circuit

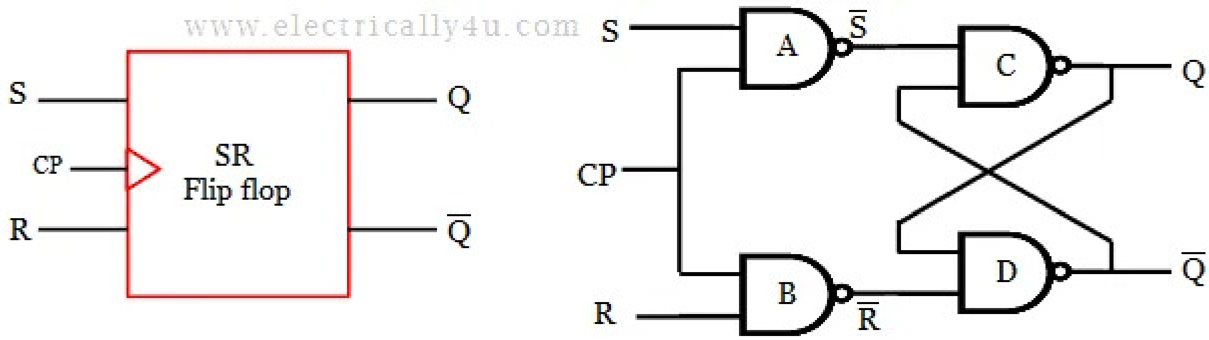
S	R	Q	$\overline{Q}$
0	0	No change	
0	1	0	1
1	0	1	0
1	1	X	X
		(Invalid)	



NOR Circuit

S	R	Q	$\overline{Q}$
0	0	No change	
0	1	1	0
1	0	0	1
1	1	X	X
		(Invalid)	

S-R Flip Flop



Clocked SR flip flop

Courtesy www.electrically4u.com

Truth Table

Describe the above in Verilog HDL and capture the Waveforms

1. ***Dataflow modeling***
2. ***Behavior modeling***